STACKS, DISCRIMINANTS: LEMMA-AGREE-DEDEKIND

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3268-3281 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 3283-3321, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 13 of queue_discriminant.json (pb=1585, stmt 3268-3281, proof 3283-3321).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0BT6: sources/stacks-discriminant/pinned-original-source.tex, source lines 821–836. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVB: sources/stacks-discriminant/pinned-original-source.tex, source lines 414–416. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BW2: sources/stacks-discriminant/pinned-original-source.tex, source lines 1695–1699. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BW1: sources/stacks-discriminant/pinned-original-source.tex, source lines 1657–1665. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVD: sources/stacks-discriminant/pinned-original-source.tex, source lines 509–513. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BWL: sources/stacks-discriminant/pinned-original-source.tex, source lines 3188–3197. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BWM: sources/stacks-discriminant/pinned-original-source.tex, source lines 3246–3255. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Assume the Dedekind different is defined for $A \to B$. Set $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. The generalization of Remark 0BWM applies to the morphism $f : Y \to X$ if and only if $1 \in \mathcal{L}_{B/A}$ (e.g., if A is normal, see Lemma 0BW1). In this case $\mathfrak{D}_{B/A}$ is an ideal of B and we have*

$$\mathfrak{D}_f = \widetilde{\mathfrak{D}_{B/A}}$$

as coherent ideal sheaves on Y.

Proof. As the Dedekind different for $A \to B$ is defined we can apply Lemma 0BWL to see that $Y \to X$ satisfies condition (1) of Remark 0BWM. Recall that there is a canonical isomorphism $c : \mathcal{L}_{B/A} \to \omega_{B/A}$, see Lemma 0BW2. Let $K = Q(A)$ and $L = K \otimes_A B$ as above. By construction the map c fits into a commutative diagram

$$\begin{array}{ccc} \mathcal{L}_{B/A} & \longrightarrow & L \\ {\scriptstyle c}\downarrow & & \downarrow \\ \omega_{B/A} & \longrightarrow & \mathrm{Hom}_K(L, K) \end{array}$$

where the right vertical arrow sends $x \in L$ to the map $y \mapsto \mathrm{Trace}_{L/K}(xy)$ and the lower horizontal arrow is the base change map (0BVB) for $\omega_{B/A}$. We can factor the lower horizontal map as

$$\omega_{B/A} = \Gamma(Y, \omega_{Y/X}) \to \Gamma(V, \omega_{V/X}) \to \mathrm{Hom}_K(L, K)$$

Since all associated points of $\omega_{V/X}$ map to associated primes of A (Lemma 0BVD) we see that the second map is injective. The element $\tau_{V/X}$ maps to $\mathrm{Trace}_{L/K}$ in $\mathrm{Hom}_K(L, K)$ by the very definition of trace elements (Definition 0BT6). Thus τ as in condition (2) of Remark 0BWM exists if and only if $1 \in \mathcal{L}_{B/A}$ and then $\tau = c(1)$. In this case, by Lemma 0BW1 we see that $\mathfrak{D}_{B/A} \subset B$. Finally, the agreement of $\mathfrak{D}_f$ with $\mathfrak{D}_{B/A}$ is immediate from the definitions and the fact $\tau = c(1)$ seen above. □